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Assessment of workforce availability, staffing, and education, and training requirements for radiographers across Europe as part of EU-REST project

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ABSTRACT

Introduction: The European Union Radiation, Education, Staffing and Training (EU-REST) project assessed workforce availability of professionals involved in medical applications of ionising radiation across European Union (EU) Member States. It developed guidelines for staffing, education and training to ensure radiation safety and quality in medical practices. This paper focuses on the current status of workforce availability, education and training requirements of radiographers across the 27 EU Member States.
Methods: A multi-phase survey was employed, consisting of a pre-survey and a main survey. The pre-survey gathered information from relevant national authorities and professional bodies, followed by the main survey distributed to these bodies to complete across all Member States. The survey covered workforce availability, staffing levels, workforce planning, education and training, and quality and safety. Data were cleaned and analysed to focus on radiographers' workforce statistics, education and training in countries that completed the relevant section of the survey.
Results: The study found significant variation in radiographer workforce availability, with a total of 171,306 working radiographers across Europe. The number of radiographers per million inhabitants ranged from 86 (Belgium) to 613 (Finland). About 7 % of radiographers were expected to retire in the next five years, with some countries facing higher projections. Variations in education systems, workforce distribution, and radiation protection practices were observed.
Conclusion: There is a need for harmonisation of how workforce data is collected and reported across Member States. Establishing centralised registries and standardised data collection methods is crucial for improving workforce management and ensuring radiation safety in medical settings.
Implications for practice: Unless coordinated actions are taken to address issues related to the radiography workforce, staffing levels, and associated education and training, optimal standards will remain unattainable.

Introduction

The European Union Radiation, Education, Staffing and Training (EU-REST) project was undertaken over a 2-year period from September 2022 by a consortium led by the European Society of Radiology (ESR), and involving the European Federation of Radiographer Societies (EFRS),[1] following a tender issued by the European Health and Digital Executive Agency (HaDEA), acting under a mandate from the European Commission's Directorate General for Health and Food Safety (DG SANTE), in collaboration with the Directorate General for Energy (DG ENER).

The primary aims of the project were to assess workforce availability (staffing levels) and education and training requirements to ensure the quality and safety of medical applications involving ionising radiation across the European Union (EU). This included the development of guidelines for staffing and for education and training for key professional groups, including radiographers, involved in maintaining radiation safety and the quality of medical radiation applications in EU Member States. A project objective was to collect and analyse data on workforce availability, education, and training needs related to the quality and safety of medical applications involving ionising radiation.

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These aims and objectives were grounded in the recognition of increasing variability in radiological workforce capacity and education standards across EU Member States, which can pose significant risks to patient safety, diagnostic and treatment accuracy, and radiation protection practices.

Concerns around safe practice have been raised in numerous contexts. Variability in radiographer staffing levels has been associated with longer waiting times, increased workloads, and reduced capacity for quality assurance and patient interaction, all of which may compromise the safe delivery of imaging and therapy services. Inadequate training in radiation protection principles has also been identified as a contributing factor to procedural errors and unnecessary radiation exposure, particularly in high-dose modalities.

The work addressed the main staff categories defined under the Basic Safety Standards Directive (BSSD), including ‘practitioner’ (radiologists, radiation oncologists, nuclear medicine physicians, and radiographers in some jurisdictions), ‘medical physics expert’ (medical physicists), and personnel responsible for the practical aspects of medical radiological procedures (radiographers), as well as those involved in reporting and learning from adverse radiological events (Table 1).[2]

For the purposes of this publication, the focus is on the category of radiographers (other professionals participating in this project has been reported to HaDEA[3] and may be published separately). Radiographers play a key role in diagnostic and interventional radiology (including ultrasound when performed by radiographers), nuclear medicine, and radiation therapy/radiotherapy/radiation oncology, depending on the classification of this group in each respective country.[3] For the EU-REST project, radiographers were defined as follows:

Radiographers, known by a variety of terms, including ‘Technologists’, working in the fields of:

i) Diagnostic and Interventional Radiology (including ultrasound, where this is performed by Radiographers)
ii) Nuclear Medicine
iii) Radiation Therapists/Radiotherapy/Radiation Oncology (where these workers fall under the category of Radiographers)

Methods

This study targeted all 27 EU Member States with the methodology for the data collection and analysis was developed and approved by HaDEA, in March 2023.

Pre-survey

Prior to the launch of the main survey, a pre-survey was conducted directed at national professional societies and associations within each EU country. Its purpose was to identify organisations capable of providing information on workforce availability, and education and training in the medical use of ionising radiation, in order to pilot the content, assess clarity, test relevance across diverse national contexts, and identify any potential ambiguities or translation issues. Feedback from the pre-survey informed refinements to survey design and terminology, helping to ensure consistency and accuracy once the main survey was disseminated. Each organisation involved in the main distribution contributed to both data collection and contextual interpretation, offering insights specific to their national structures and workforce challenges.

The European-level professional societies involved in the project: European Society of Radiology (ESR), European Society for Therapeutic Radiology and Oncology (ESTRO), European Federation of Organisations for Medical Physicists (EFOMP), European Federation of Radiography Societies (EFRS), and European Association of Nuclear Medicine (EANM) (as an advisory board member) maintain up-to-date databases of national professional societies related to the relevant professional categories Based on these databases, a pre-survey was disseminated in English via SurveyMonkey (SurveyMonkey Inc., San Mateo, USA) to collect information on national authorities and professional bodies across the EU27 countries responsible for staffing, education, and training. The survey design supported both structured responses (e.g., dropdowns) and free-text inputs.

The pre-survey yielded 273 contacts from relevant authorities and professional bodies in the EU27, creating a comprehensive database of relevant contacts for all Member States for the distribution of the Main Survey.

The methodology encompassed several key elements to comprehensively assess staffing and training related to the medical use of ionising radiation. First, the main professional groups involved were clearly defined. This was followed by an evaluation of staff availability across these groups. The survey then examined their professional qualifications and relevant training. Guidance documents on safe and high-quality radiation practices were reviewed. Data collection involved selecting appropriate sources and methods. Finally, a structured approach to data analysis was established to support both EU-wide and country-specific evaluations.

Table 1
Main staff categories defined under the BSSD.[2]

Main staff categories	Description	Profession/s
Practitioner	A medical doctor, dentist or other health professional who is entitled to take clinical responsibility for an individual medical exposure in accordance with national requirements.	Radiologists, radiation oncologists, nuclear medicine physicians, and radiographers in some jurisdictions
Medical physics expert	An individual or, if provided for in national legislation, a group of individuals, having the knowledge, training and experience to act or give advice on matters relating to radiation physics applied to medical exposure, whose competence in this respect is recognised by the competent authority.	Medical physicist
Personnel responsible for the practical aspects of medical radiological procedures	Personnel responsible for the physical conduct of a medical exposure and any supporting aspects, including handling and use of medical radiological equipment, the assessment of technical and physical parameters (including radiation doses), calibration and maintenance of equipment, preparation and administration of radiopharmaceuticals, and image processing.	Radiographers

Main Survey

The main survey was designed to gather detailed information relevant to the project's core objectives: assessing workforce availability, education and training standards, planning strategies, and the quality and safety of radiological practice across EU Member States. The survey was structured into four thematic sections:

- Section A – Education and Training: This section explored the qualifications, initial education programmes, continuing professional development (CPD) requirements, and specialist training opportunities available to radiographers in each country.
- Section B – Workforce Availability: This section collected quantitative data on the number of practising radiographers, vacancy rates, recruitment challenges, and demographic details such as age distribution and international recruitment.
- Section C – Workforce Planning: This section assessed whether national workforce planning strategies existed, and if so, what data sources and forecasting models were used to plan for future radiography workforce needs.
- Section D – Quality and Safety: This section evaluated systems in place to ensure quality and safety in radiological services, including adherence to radiation protection standards, audit procedures, and mechanisms for incident reporting and learning.

While the full survey is provided in the supplementary materials (Supplementary materials 1), this summary is included here to provide context for the alignment between the survey content and the project's objectives of promoting harmonisation, safety, and sustainability in medical radiation services.

The main survey was distributed to the key national stakeholders including:

- all national organisations and authorities from the database established through the pre-survey
- all EU27 national professional societies for radiology/nuclear medicine/radiotherapy/radiographers/medical physics through ESR, EANM, ESTRO, EFRS and EFOMP
- all EU27 national radiation protection authorities through Heads of the European Union Radiological Protection Competent Authorities (HERCA)
- all EU27 national medical associations/chambers through European Union of Medical Specialists (UEMS)

The decision to involve multiple professional bodies in distributing the main survey was intentional and strategic. It aimed to maximise response rates, ensure geographical and professional representativeness, and facilitate access to accurate workforce data, which is often held or reported differently across countries. These organisations also provided essential contextual knowledge about national education and regulatory frameworks, which was critical to interpreting responses accurately.

Like the pre-survey, the main survey was conducted in English using SurveyMonkey.

Data cleaning

Data cleaning aimed to ensure consistency rather than verify accuracy. When multiple responses were received from a given source (e.g., national authority or society), the most complete response was selected. Additional responses from the same professional category within the same country were compared for consistency. Substantial discrepancies were flagged and clarified with the relevant stakeholders. Ultimately, one validated response per professional category and national society was retained for each country. Where available, corresponding responses from national authorities were also included to explore potential differences. Full methodological details are provided in the official European Commission (EC) EU-REST report.[3]

Results

Pre-survey

A total of 109 responses were received, including at least one from each EU27 country. All the targeted professional groups (radiologists, medical physicists, nuclear medicine physicians, radiation oncologists, and radiographers) provided responses (see Table 2), although not all were represented in every EU27 country. Radiographers had a response rate of 20.2 %. The radiographer response rate in the pre-survey does not limit the findings reported in this paper, which are based on data collected from the main survey where radiographers were the primary target group. Nonetheless, the input from radiographers in the pre-survey phase was valuable in identifying discipline-specific terminology and ensuring the final survey instrument was appropriately tailored for the main study.

Following an analysis of the incomplete responses, appropriate follow-up actions were taken, such as contacting national societies to verify or complete the missing information. At the end of this process a total of 186 responses of various levels of completeness were received.

Main Survey

The exact number of survey invitations circulated is unknown, as distribution was managed independently by multiple national and professional organisations. This decentralised approach was intended to maximise reach and engagement across all EU27 Member States but limited our ability to track distribution volume and calculate a precise response rate. Despite this, the survey initially received 101 responses related to radiographers from 21 of the 27 EU Member States. Following an extension period, the total number of responses increased with representation expanding to 23 Member States. After data cleaning (removing incomplete or duplicate entries) valid full responses were retained from 20 EU Member States. Despite these adjustments, the dataset provided broad geographic coverage and supports the reliability and relevance of the findings.

The results from the main survey, after data cleaning will be presented for radiographers, in line with the specifications outlined in the EU HaDEA tender.[4]

Workforce/staffing

According to the main survey initial results, there were 171,306 working radiographers reflecting the workforce across the 27 EU

Table 2
Pre-Survey responses by professional groups.

Group	Responses % (n)
Medical Doctor (radiologist, radiation oncologist, and/or nuclear medicine physician)	45.9 % (50)
Radiographer/Radiation Therapist	20.2 % (22)
Medical physicist	29.4 % (32)
Other profession using ionising radiation (focusing on high-dose procedures)	4.6 % (5)

member states at the time of the survey (2022–2023). The EU average radiographer workforce was a ratio of 385 radiographers per million inhabitants while the ratio ranged from 86 radiographers per million inhabitants in Belgium (BE) up to 613 radiographers per million in Finland (FI) (see Fig. 1). The accompanying colour map (see Fig. 2) illustrates the geographical distribution of radiographers across the EU27, highlighting the 13 countries with a density lower than the EU average (depicted in dark orange to yellow) and the 10 countries with a higher density (depicted in light to dark green). Data from Cyprus (CY), Hungary (HU), and Romania (RO) were missing.

In terms of workforce availability for the 17 countries that provided age profile data (Table 3), approximately 7 % (10,270) of radiographers were expected to retire within the next 5 years, and 30 % were over the age of 51 years. Eleven countries, the Czech Republic (CZ), Denmark (DK), Estonia (EE), Finland (FI), Greece (GR), Italy (IT), Latvia (LV), Lithuania (LT), Poland (PL), and Slovenia (SI) are projected to lose a higher percentage of their radiographer workforce to retirement in the next 5 years than the EU average of 7 %, based on a retirement age of 66 years. Poland and Slovenia have the highest projected losses (at 20 %).

Workforce overview in Europe

Radiographers were by far the largest group (67 %), followed by radiologists (24 %), medical physicists (4 %), radiation oncologists (3 %) and nuclear medicine physicians (2 %).

As part of an analysis of workforce distribution, five EU countries with approximately the same population (10 million) were compared: Czech Republic (CZ), Greece (GR), Hungary (HU), Portugal (PT), and Sweden (SE) (Table 4).

The table highlighted significant heterogeneity in the availability of health professionals involved in the use of ionising radiation across these five countries. For radiographers, the maximum difference was a factor of 1.7 (3000 for GR and 5000 for PT), making radiographers the professional group with the smallest variation.

Table 5 presents workforce distribution (per 1,000,000 inhabitants) to facilitate comparison between countries, with the EU average, median, minimum, maximum, and difference factor (the ratio between the maximum and minimum values). The highlighted cells indicate values lower than the EU average i.e. 12 countries for radiographers (versus 16 countries for radiologists, 16 for radiation oncologists, 15 for nuclear medicine physicians, and 13 for medical physicists).

Education and training

Any type of education and training in radiation protection during pre-registration programmes varied from 2 to 52 weeks, and the majority of countries (n = 11) required specific certification in radiation protection. Furthermore, 19 countries mandated CPD for radiographers (see Table 6).

Discussion

This study was the first to assess the workforce availability for health professionals involved in ionising radiation-based diagnostic and therapeutic procedures, such as radiographers, and combines this with gathering data on the education and training, specifically linked to radiation protection.[1] The results revealed significant heterogeneity across Member States and professional groups, which can impact healthcare delivery and the competence of professionals in radiation protection.

This study underscores the need for clear guidance and standardised metrics on radiographic workforce availability to

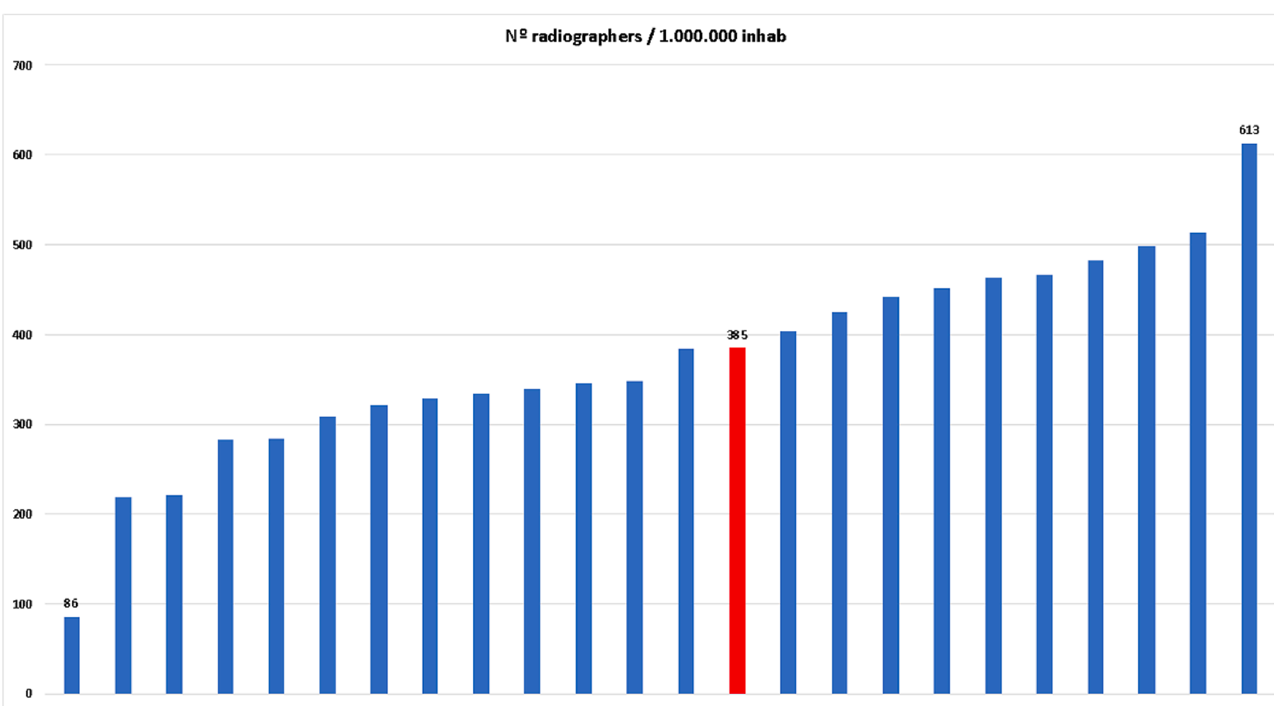

Figure 1. Number of radiographers per million inhabitants *(includes estimated workforce numbers provided by Member States who went on to submit an incomplete response overall i.e. 'not valid/full')* Key: AT (Austria); BE (Belgium); BG (Bulgaria); HR (Croatia); CY (Cyprus); CZ (Czechia); DK (Denmark); EE (Estonia); FI (Finland); FR (France); DE (Germany); HU (Hungary); IE (Ireland); IT (Italy); LV (Latvia); LT (Lithuania); MT (Malta); NL (Netherlands); PL (Poland); PT (Portugal); RO (Romania); SK (Slovakia); SI (Slovenia); ES (Spain); SE (Sweden).

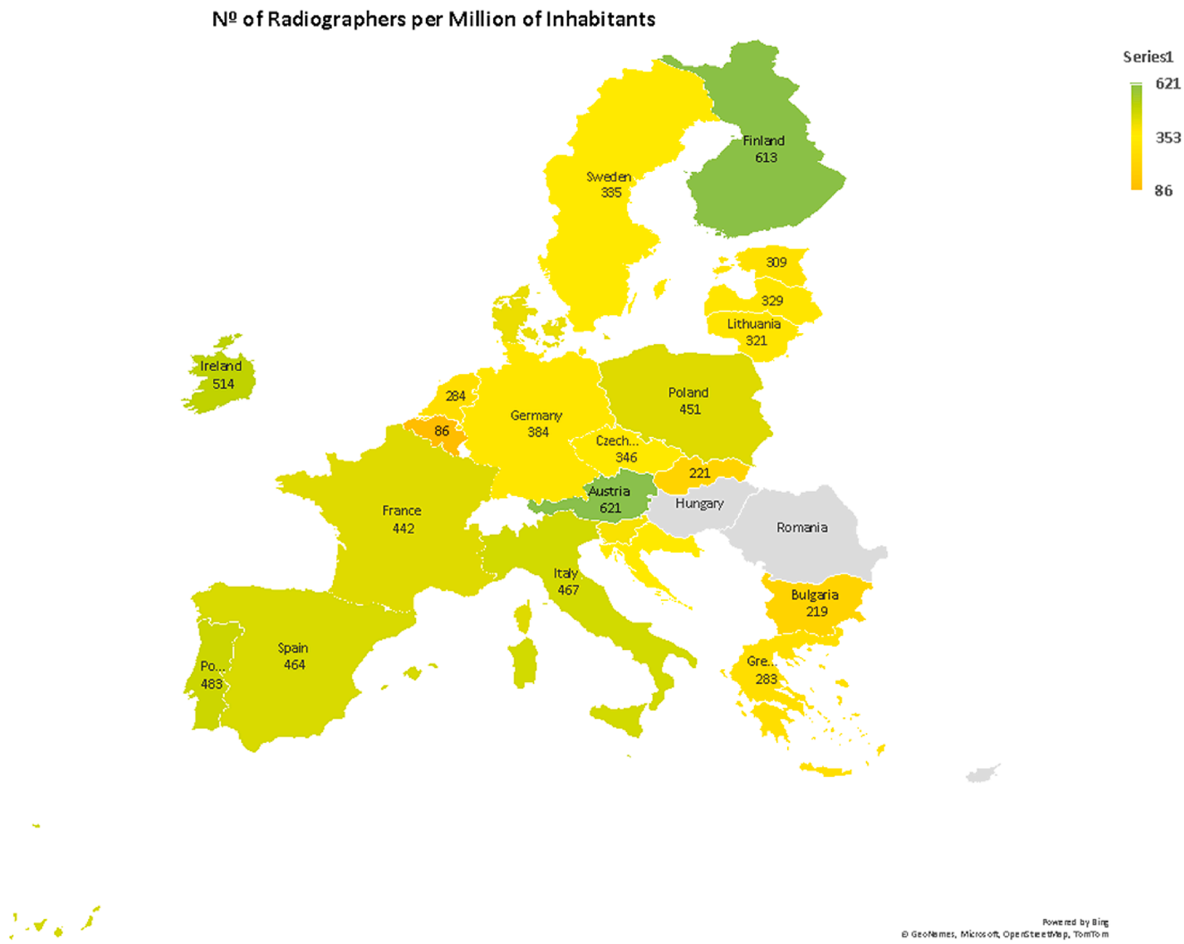

Figure 2. Geographical distribution of radiographers *(EU average = 385 radiographers* per *million inhabitants).*

Table 3
Radiographers' age profile based on data provided by 17 countries.

Country	No. due to retire in 5 years (%)	≤50 years old (%)	≥51 years old (%)
Austria	446 (8 %)	72 %	28 %
Czechia	510 (14 %)	61 %	39 %
Denmark	250 (10 %)	70 %	30 %
Estonia	62 (15 %)	71 %	29 %
Finland	578 (17 %)	66 %	34 %
France	0 (0 %)*	76 %	24 %
Germany	1600 (5 %)	80 %	20 %
Greece	300 (10 %)	50 %	50 %
Ireland	130 (5 %)	85 %	15 %
Italy	2240 (8 %)	72 %	28 %
Latvia	62 (10 %)	65 %	35 %
Lithuania	90 (10 %)	50 %	50 %
Malta	8 (3 %)	90 %	10 %
Poland	3400 (20 %)	65 %	35 %
Portugal	250 (5 %)	85 %	15 %
Slovenia	170 (20 %)	60 %	40 %
Sweden	175 (5 %)	80 %	20 %
17 countries combined	**10,270 (7 %)**	**70 %**	**30 %**

** While a response was provided from France, a figure of '0' was provided for the number of radiographers due to retire in 5 years.*

ensure equitable patient access and improve healthcare quality across Europe. While the current analysis provides important insights into workforce staffing levels, a more comprehensive understanding could be achieved by correlating these findings with healthcare performance indicators. For instance, regions with fewer radiographers might also experience longer wait times or delays in diagnosis or treatment for high-priority conditions such as cancer. Additional data on patient satisfaction, staff retention, and broader healthcare quality measures could provide a better perspective on the systemic impact of workforce variability. Currently, the absence of such standards complicates the identification of best practices.

Table 4
Workforce distribution in 5 countries with similar population *(approx. 10 million)*.

Country	Radiographers	Radiologists	Medical Physicists	Radiation Oncologists	Nuclear Medicine Physicians
Czechia	3643	1321	153	276	164
Greece	3000	2800	350	28	208
Hungary	No data	700	80	100	90
Portugal	5000	987	120	130	55
Sweden	3500	2820	450	No data	244

Table 5
Workforce distribution per 1 million inhabitants.

Country	Radiographers	Radiologists	Medical Physicists	Radiation Oncologists	Nuclear Medicine Physicians
Austria	621	161	27	No data	18
Belgium	86	138	21	16	36
Bulgaria	219	51	12	11	6
Croatia	348	150	17	40	18
Cyprus	No data	144	24	14	10
Czechia	346	126	15	26	16
Denmark	426	115	26	32	21
Estonia	309	154	20	38	8
Finland	613	116	27	41	20
France	442	131	13	14	10
Germany	384	115	36	18	15
Greece	283	264	33	3	20
Hungary	No data	72	8	10	9
Ireland	514	73	32	6	2
Italy	467	233	20	17	25
Latvia	329	133	16	No data	4
Lithuania	321	107	4	16	No data
Malta	499	77	35	15	6
Netherlands	284	72	24	16	11
Poland	451	112	10	17	5
Portugal	483	95	12	13	5
Romania	No data	105	13	8	4
Slovakia	221	74	22	15	8
Slovenia	403	119	15	23	8
Spain	464	105	25	21	9
Sweden	335	270	43	No data	23
EU average	**385**	**127**	**21**	**19**	**13**
No. of countries below EU average	**12**	**16**	**13**	**16**	**15**
EU median	**384**	**115**	**21**	**16**	**10**
No. of countries below EU median	**11**	**13**	**13**	**11**	**11**
Min.	86.0	51.2	4.3	2.6	2.0
Max.	621.0	269.8	43.1	41.5	35.5
Difference factor	**7.2**	**5.3**	**10.1**	**15.7**	**18.0**

Table 6
Training requirements for radiographers.

Country	Pre-registration Education Programme Duration (years)	Radiation Protection Training (weeks)	Specific Radiation Protection Certification required	Mandatory CPD in Radiation Protection
Austria	3	16–24	No	Yes
Belgium	3	No data	No data	No data
Bulgaria	3	28–52	Yes	Yes
Croatia	3	16–24	Yes	Yes
Cyprus	No data	No data	No data	No data
Czechia	3	No data	Yes	Yes
Denmark	3.5	2–4	No	Yes
Estonia	3.5	52	No	Yes
Finland	3	No data	No	Yes
France	3	2–4	Yes	Yes
Germany	3	16–24	Yes	Yes
Greece	4	16–24	Yes	No
Hungary	No data	No data	No data	No data
Ireland	4	No data	No data	Yes
Italy	3	<2	No	Yes
Latvia	3	52	Yes	Yes
Lithuania	3	16–24	No	Yes
Malta	4	52	No	Yes
Netherlands	4	No data	No data	No data
Poland	2.5	No data	Yes	Yes
Portugal	4	52	No	Yes
Romania	No data	No data	No data	No data
Slovakia	3	No data	No	Yes
Slovenia	3	4–12	Yes	Yes
Spain	2	No data	Yes	Yes
Sweden	3	52	Yes	Yes

Workforce

The variation in radiographers per million inhabitants could be explained through the differing stages of development in education and training programmes as well as the varying levels of recognition of radiographers as a profession across Member States. In some countries, the establishment of formal education and training for radiographers has only occurred relatively recently. As a result, in these countries, other healthcare professionals, such as nurses or medical technologists, may have assumed the roles traditionally occupied by radiographers in other countries, leading to a smaller, less specialised workforce in the field. The variation in the population of radiographers in the five countries with population of approximately 10 million (Table 4) highlights the large difference, by a factor of 1.7, between the 3000 radiographers in Greece and the 5000 in Portugal. However, this variance was much less than those seen for radiologists (a factor of 4 between Hungary (n = 700) and Sweden (n = 2820)), medical physicists (a factor of 5.6 between Hungary (n = 80) and Sweden (n = 450)), radiation oncologists (a factor of 9.9 between Greece (n = 28) and Czechia (n = 276)), and nuclear medicine physicians (by a factor of 4.4 between Portugal (n = 55) and Sweden (n = 244)). While differences in healthcare system organisation may partly explain these discrepancies, other contributing factors are likely. Exploring associations with additional datasets could offer further insights. However, such analyses were beyond the scope of the current study and should be considered in future research.

In terms of workforce availability, retirement projections over the next five years showed that approximately 7 % of the radiographer workforce were due to retire (from the 17 countries that provided retirement data; Table 2); however, for Czechia (14 %), Estonia (15 %), Finland (17 %), Poland (20 %), Slovenia (20 %) the projected retirements over the next five years were at least double the average. With an average retirement age in the EU of 61.3 years,[5] 4.7 years lower than the 66 years used in this study, the proportion of radiographers who were 51 years or older is concerning, with eight of the 17 countries reporting between 30 % and 50 % of the workforce in this category (Table 3).

The EFRS member surveys traditionally ask national societies whether there are sufficient jobs for all graduating radiographers entering the workforce each year. In their 2020 survey,[6] for medical imaging, 19 national societies indicated that there were enough vacancies, whereas six indicated that there were not enough vacancies for graduates in their country. For radiotherapy, 18 indicated that there were enough jobs while six again indicated that there were not enough jobs. Finally, for nuclear medicine, 17 indicated that there were enough jobs while seven indicated that there were not enough jobs for graduates. Unfortunately, in collecting the workforce data, the current survey did not capture information on the specific fields in which radiographers are employed, such as radiology, nuclear medicine, or radiotherapy. This represents an important gap, as the roles and responsibilities of radiographers vary significantly across these domains, with implications for workforce planning, education, and retention strategies Some jurisdictions, such as the UK, which was not included in the present study, routinely publish detailed workforce reports.[7,8] For example, the UK's 2022 report[7] on the medical imaging radiographer workforce identified a vacancy rate of 12.8 %, with 4.7 % of the workforce aged over 60 years and 8.1 % recruited internationally. For radiotherapy radiographers, the vacancy rate was slightly lower at 8.3 %, with 5 % expected to retire within three years, and 2.4 % recruited internationally.[8] These data suggest workforce instability and a growing reliance on international recruitment, trends that may also be present in other European countries included in our study but remain unquantified due to data limitations. The broader literature reinforces this concern. In their 2024 commentary *"The shortage of radiographers: a global crisis in healthcare"*, Konstantinidis[9] highlights the urgent need to address global workforce deficits, while similar challenges have been documented in the United States.[10,11] These external data sources collectively underscore the importance of capturing more granular workforce information in future European surveys. Doing so would enable more accurate identification of at-risk areas, inform targeted workforce strategies, and support international comparisons. Therefore, while the current study offers a foundational overview, there is a pressing need for more detailed, harmonised workforce reporting across Europe to address this growing challenge proactively.

Education, training and radiation protection

The length of training often correlates with the structure and recognition of these programmes within national educational systems. In some countries, such as Spain (ES) and Germany (DE), the education and training programmes for radiographers remain unintegrated to the higher education system, which may limit the recognition and standardisation of qualifications across Europe. In their 2020 member survey,[6] the EFRS reported that 13 national societies (44.8 % of responding societies) had vocational level education and training at either European Qualifications Framework (EQF) level 6 (i.e. equivalent to Bachelors) or level 5. This suggested more vocational programmes outside of the higher education system than in previous member surveys[12]; however, this data was likely impacted by a lower response rate from societies as a more recent international study found that 84 % of programmes were Bachelor's level and less than five were vocational at EQF levels 5 or 6.[13] In the cases of some countries, such as Spain,[14] the level of education and training of radiographers persist as 2 year vocational programmes, despite the ongoing campaign of national societies and the EFRS. Others have highlighted the lack of regulation of training programmes for radiotherapy leading to underdeveloped professional competencies and thus potential deficits in patient care.[15]

The European Directive 2013/51/EURATOM requires adequate education and training in radiation protection for radiographers; however, the implementation varies greatly among Member States,[1] despite the past guidance provided by the Medical Radiation Protection Education and Training (MEDRAPET) project.[16] The European Medical Application and Radiation Protection Concept: Strategic Research Agenda and Roadmap Interlinking Health and Digitisation Aspects (EURAMED rocc-n-roll) project also highlighted the variation in radiation protection education and training across professions and across Europe.[17] Large variations with compliance with the BSSD, with deficits in education and training across professions at undergraduate and postgraduate levels, and with perceptions on the problems leading to these deficits were highlighted in their survey.[17] EURAMED rocc-n-roll also explored the importance of education and training in radiation protection, noting a lack of effective application of these principles in practice as a key challenge.[18] Success in this area depends on strong governance, leadership, and financial support, as well as the creation of a sustainable, pan-European radiation protection training network. This was also reflected in an International Atomic Energy Agency (IAEA) project which discussed the heterogeneity of radiographer education and training, specifically linked to radiation protection, in Europe and Central Asia.[19] Duration, content, and quality of these programmes were identified as areas of concern which, when combined, likely impact on quality of patient care delivered.

The length of any type of radiation protection training for radiographers varied widely, from as little as 2 weeks up to 52 weeks (Table 6). Despite this diversity, most countries require specific certification in radiation protection, and CPD in radiation protection is mandatory in many countries. The 2016 EFRS study exploring patient safety issues across radiography curricula highlighted significant opportunities to address deficits and to further develop the teaching and assessment of topics from the introductory level and on to an advanced level in terms of radiation protection topics.[20] Studies have shown positive attitudes among radiographers towards CPD, including mandatory CPD once barriers such as access, funding, and time are overcome.[21–23] The current EU-REST project found that mandatory CPD for radiographers in radiation protection was in place in 20/27 countries and specific radiation protection certification was required in 11/27. The project has already highlighted the need for mandatory CPD in radiation protection for radiologists in another recent publication.[24]

Although radiation protection training was reported across Member States through undergraduate curricula, certification processes, or CPD, there was substantial variation in format, content, and regulatory oversight. This diversity in approach may reflect national autonomy and different clinical models; however, it presents challenges in assessing whether all radiographers, regardless of country, meet a consistent threshold of competence in radiation safety. The current study does not suggest that variation equates directly to unsafe practice, but it does raise the need for further investigation into whether shared minimum standards, such as those aligned with European BSSD requirements, could help support cross-border workforce mobility and assurance of patient safety. Defining such a framework could be a valuable next step for European stakeholders.

Limitations

This study faced several limitations common to survey-based research, including a variable and non-compulsory response rate, data not received for some sections from all Member States and potential issues arising from using a single language (English) for the survey, which may lead to misinterpretation. While efforts were made during the data cleaning phase to address these gaps, success was limited, and due to the voluntary nature of participation, no further data could be obtained for missing information, following the pre-established methodology of the study.

A key limitation of this study is the lack of linkage between workforce data and other healthcare system performance metrics, such as wait times, disease-specific diagnostic and treatment delays, patient satisfaction, and staff turnover rates. Such correlations could enrich our understanding of how workforce shortages tangibly affect service delivery and outcomes. However, these analyses were beyond the scope of the current study. Future research should explore these relationships, contingent on access to relevant datasets and appropriate ethical approvals. If permitted by the data custodians, secondary analyses using the existing dataset could offer valuable evidence to inform workforce planning and health policy at both national and European levels.

Recommendations

The EU-REST consortium recommends that Member States establish and maintain a centralised registry (preferably in digital format) to collect and update data on the workforce involved in ionising radiation-related professions. This registry should include key demographics on the workforce and would be useful for professional societies and policymakers, at national and EU levels, and hospital managers. It would also enhance the quality of data for EU reports and research. The study found that current data on these areas is difficult to obtain, unreliable, and varies greatly between countries, highlighting the need for uniform, centralised registries in the future.

Conclusion

This study provided a comprehensive analysis of the workforce availability and education and training, linked to radiation protection, for radiographers across EU Member States. The findings highlighted significant heterogeneity in staffing levels, education and training structures, and radiation protection practices-related requirements. While most countries reported that radiation protection was addressed through baseline training, certification, or CPD, the content, delivery, and assessment mechanisms varied. Although this diversity may reflect local needs and systems, it raises questions about consistency in meeting core competencies for safe practice across the EU. Further evaluation of whether existing training frameworks are sufficient to ensure consistent standards in radiation protection knowledge and skills is warranted.

The study emphasises the need for the establishment of centralised registries and better coordination in data collection to be able to understand current and future workforce management. Additionally, there is a need for the harmonisation of practices and data across EU Member States to ensure the effective and safe use of ionising radiation in medical settings, while also stressing the importance of continuous education and training in radiation protection to safeguard both patients and healthcare professionals.

Ethics approval and consent to participate

The European Commission initiated EU-REST Project did not require Institutional Review Board approval or informed consent because responses were provided by national regulators and national professional societies.

Availability of data

European Commission: European Health and Digital Executive Agency, Analysis on workforce availability, education and training needs for the quality and safety of medical applications involving ionising radiation in the EU—Status and recommendations—Final report, Publications Office of the European Union, 2025, https://doi.org/10.2925/2213975.

Author contributions

FZ, JMcN: Conceptualisation, Methodology, Validation, Formal Analysis, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualisation, Project Administration, Funding Acquisition.

Generative AI use

Not applicable.

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Conflicts of interest statement

JMcN is the current Editor in Chief at *Radiography*, however, as an author of this submission he had no role in or visibility of the handling of the manuscript through the editorial or peer review process.

The remaining authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.radi.2025.103005.

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